Project Documentation

C64 Expansion Board Extension
Port Side

Project number: 226

Revision: 0

Date: 25.05.2026

C64 Expansion Port Extension Port Side Rev. 0

Module Description

Introduction

The purpose of the C64 Expansion Port Extension is to place an Expansion Port connector at a desired location and connect the actual C64 Expansion Port via a ribbon cable. The maximum length of said ribbon cable is not yet determined. This board is connected to the C64 itself.

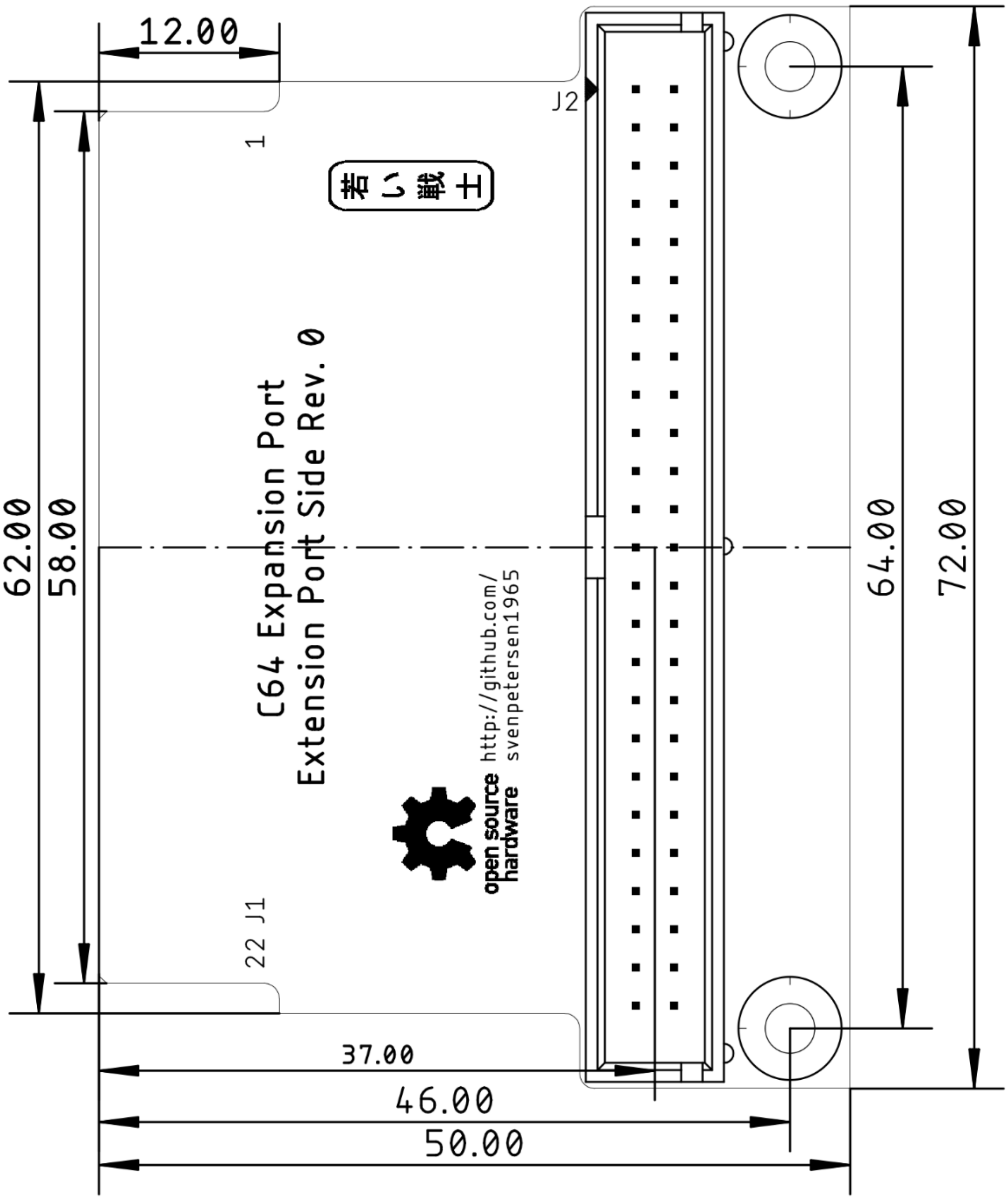


Figure 1: Dimension of the port side PCB

Connector Pinouts

J1 – Expansion Port

Pin (TOP)	Signal	Pin	Signal (BOT)
1	GND	A	GND
2	+5V	B	$\overline{\text{ROMH}}$
3	+5V	C	$\overline{\text{RESET}}$
4	$\overline{\text{IRQ}}$	D	$\overline{\text{NMI}}$
5	R / $\overline{\text{W}}$	E	PHI2
6	DOT CLOCK	F	A15
7	$\overline{\text{I/O1}}$	H	A14
8	$\overline{\text{GAME}}$	J	A13
9	$\overline{\text{EXROM}}$	K	A12
10	$\overline{\text{I/O2}}$	L	A11
11	$\overline{\text{ROML}}$	M	A10
12	BA	N	A9
13	$\overline{\text{DMA}}$	P	A8
14	D7	R	A7
15	D6	S	A6
16	D5	T	A5
17	D4	U	A4
18	D3	V	A3
19	D2	W	A2
20	D1	X	A1
21	D0	Y	A0
22	GND	Z	GND

J2 – Break out connector

2x25 box header, 2.54mm pitch

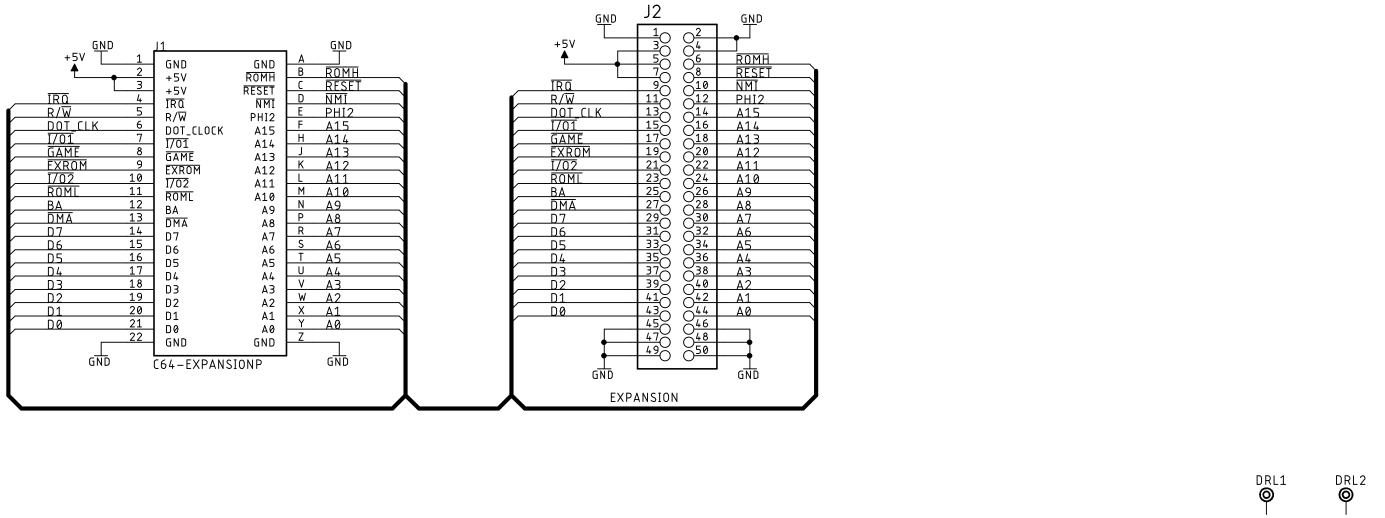
Pin	Signal	Pin	Signal
1	GND	2	GND
3	+5V	4	GND
5	+5V	6	$\overline{\text{ROMH}}$
7	+5V	8	$\overline{\text{RESET}}$
9	$\overline{\text{IRQ}}$	10	$\overline{\text{NMI}}$
11	R / $\overline{\text{W}}$	12	PHI2
13	DOT CLOCK	14	A15
15	$\overline{\text{I/O1}}$	16	A14
17	$\overline{\text{GAME}}$	18	A13
19	$\overline{\text{EXROM}}$	20	A12

Pin	Signal	Pin	Signal
21	$\overline{\text{I/O2}}$	22	A11
23	$\overline{\text{ROML}}$	24	A10
25	BA	26	A9
27	$\overline{\text{DMA}}$	28	A8
29	D7	30	A7
31	D6	32	A6
33	D5	34	A5
35	D4	36	A4
37	D3	38	A3
39	D2	40	A2
41	D1	42	A1
43	D0	44	A0
45	GND	46	GND
47	GND	48	GND
49	GND	50	GND

Revision History

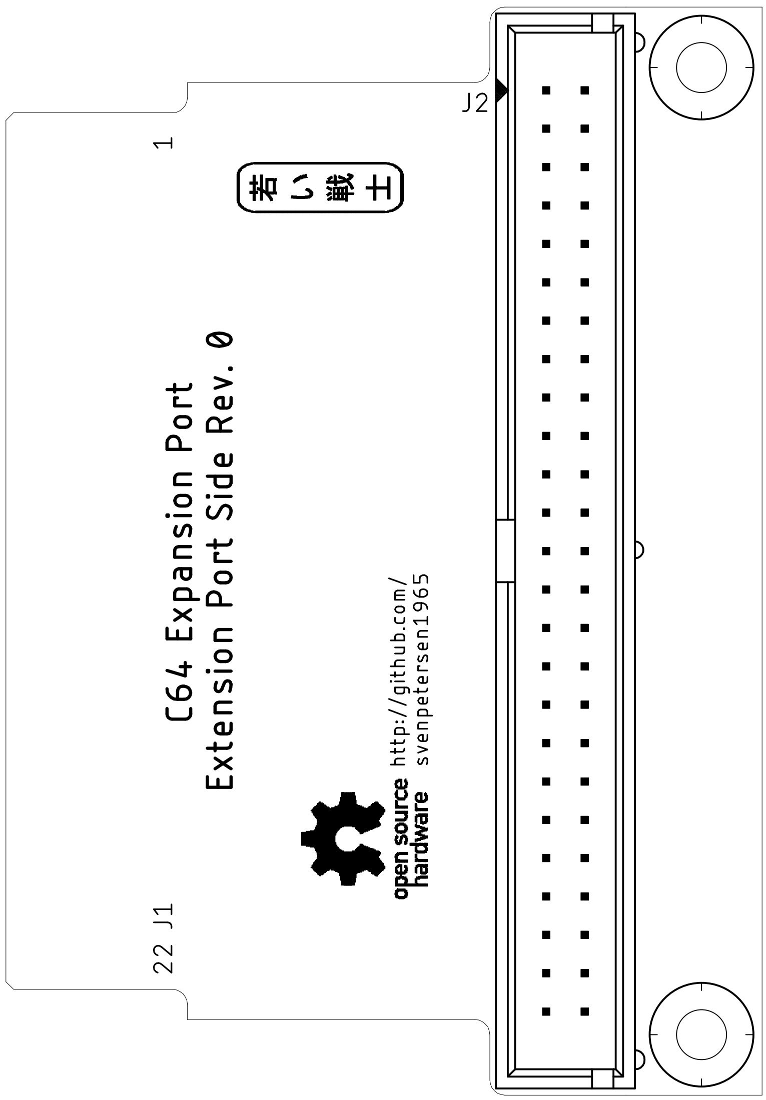
Rev. 0

The prototype is functional with a 25cm cable



Title: Break out for Expansion Port Port Side		Doc.-No.: 226-1-01-00	
		Draft.: Sven Petersen	
Date: 22.05.2026 15:33		Rev.: 0	Page 1/1
File: C64_Exp_port			
http://github.com/svenpetersen1965			A3

Sven Petersen 2026	Doc.-No.: 226-2-01-00	
	Cu: 35µm	Cu-Layers: 2
C64_Exp_port		
25.05.2026 17:58		Rev.: 0
placement component side		



Sven Petersen 2026	Doc.-No.: 226-2-01-00	
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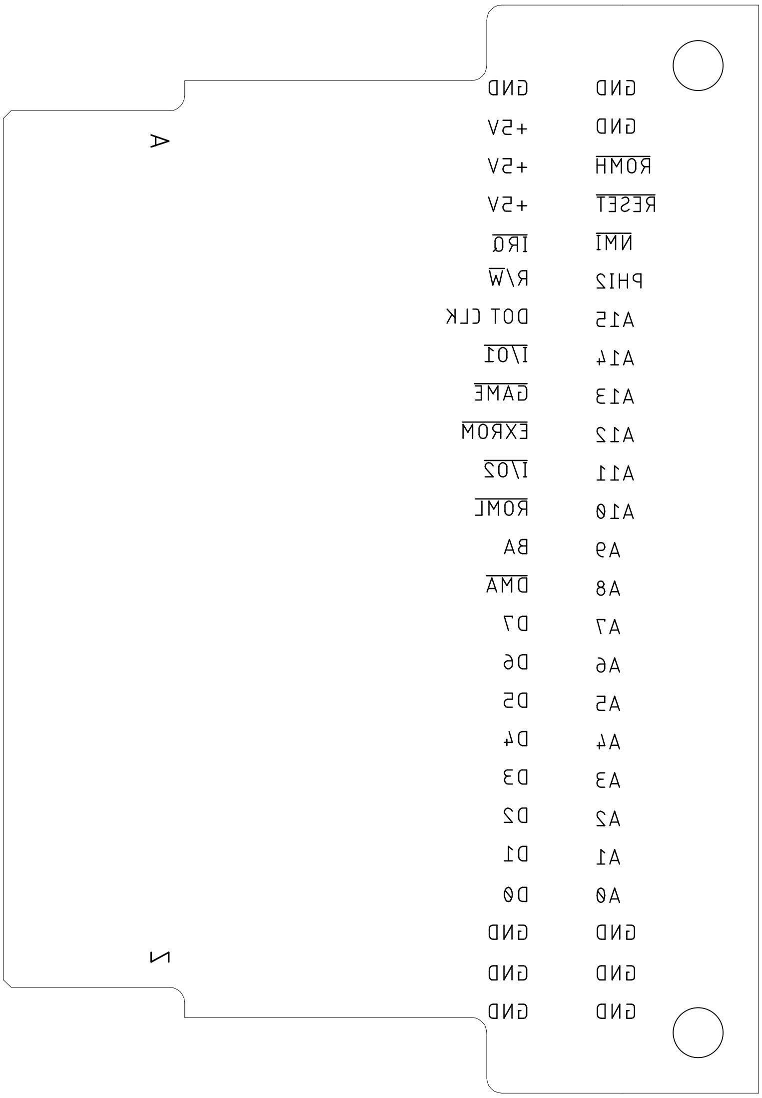
C64_Exp_port		
25.05.2026 17:58		Rev.: 0
placement solder side		



Sven Petersen 2026	Doc.-No.: 226-2-01-00	
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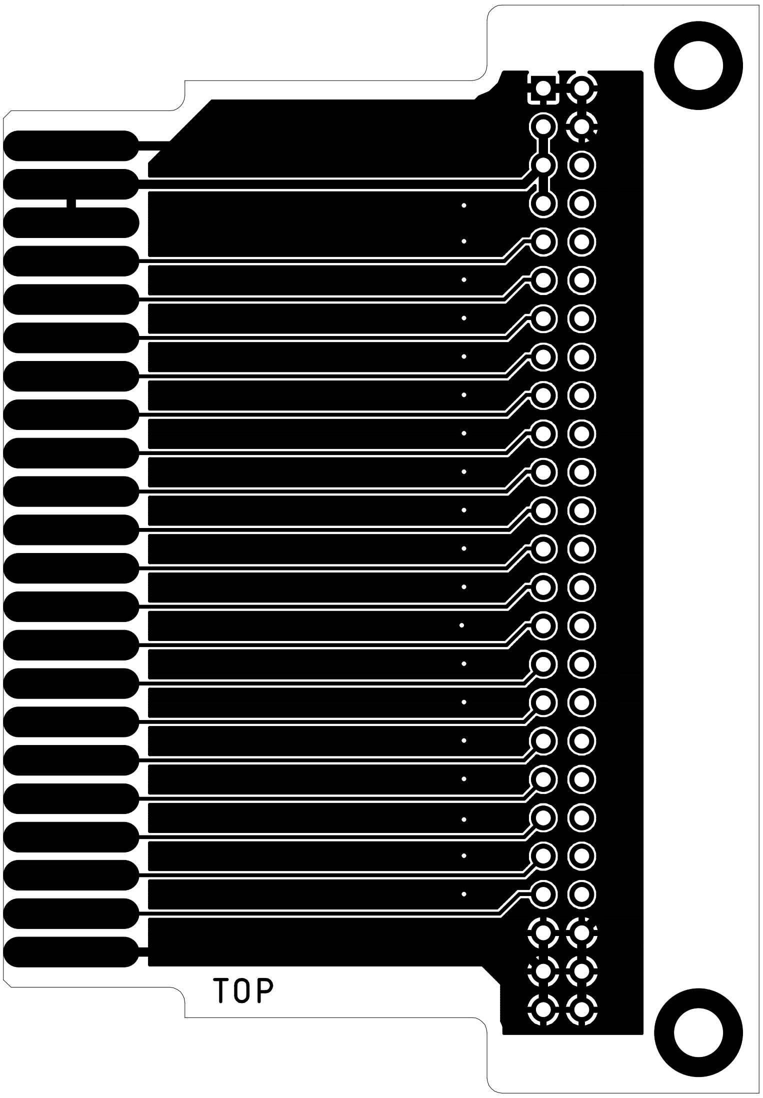
C64_Exp_port		
25.05.2026 17:58		Rev.: 0
top [redacted]		



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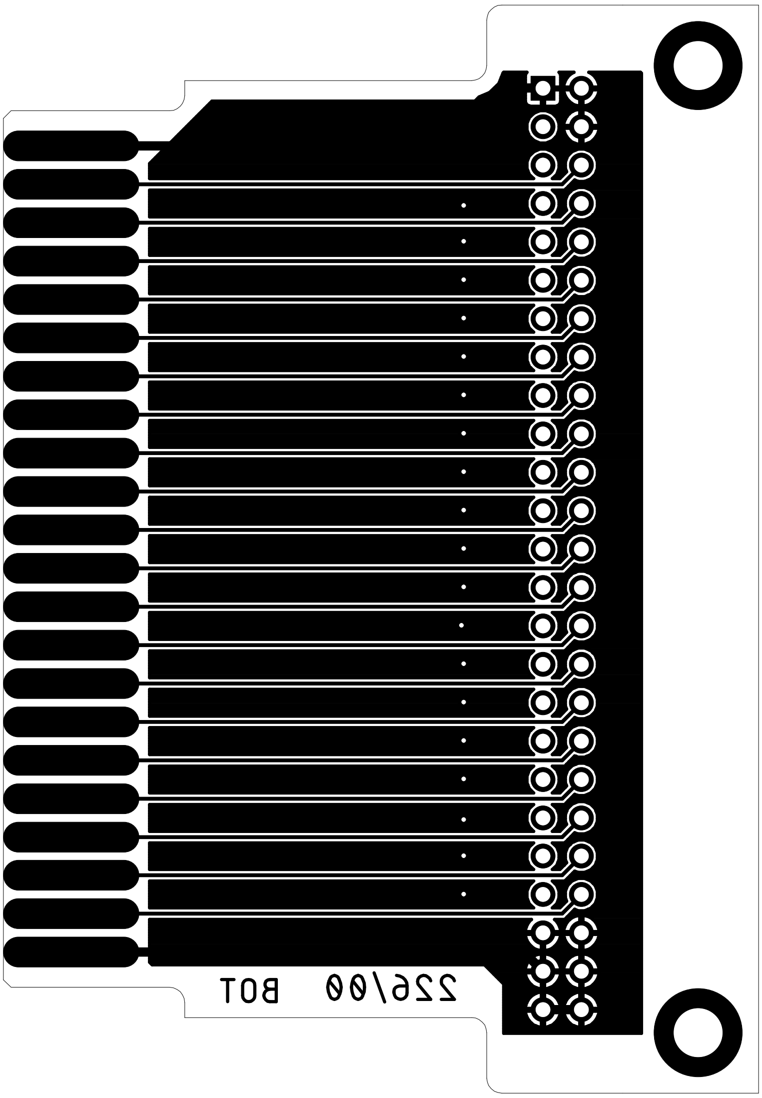
C64_Exp_port		
25.05.2026 17:58		Rev.: 0
bottom	[redacted]	



Sven Petersen 2026	Doc.-No.: 226-2-01-00	
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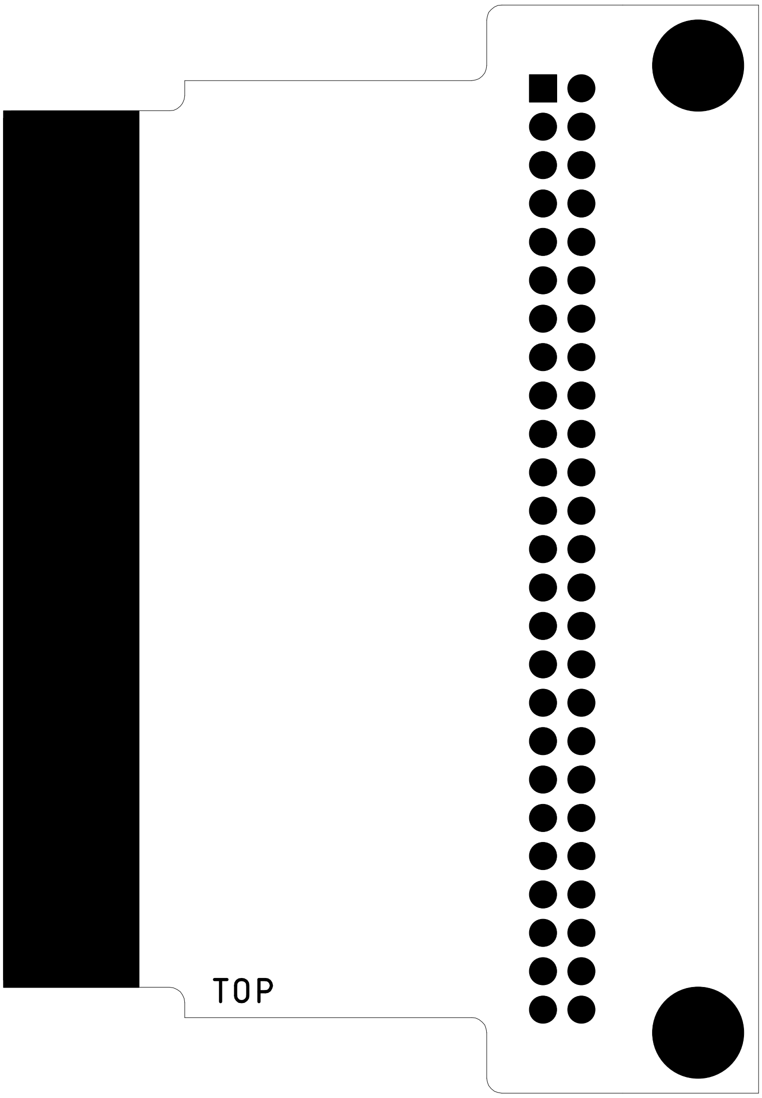
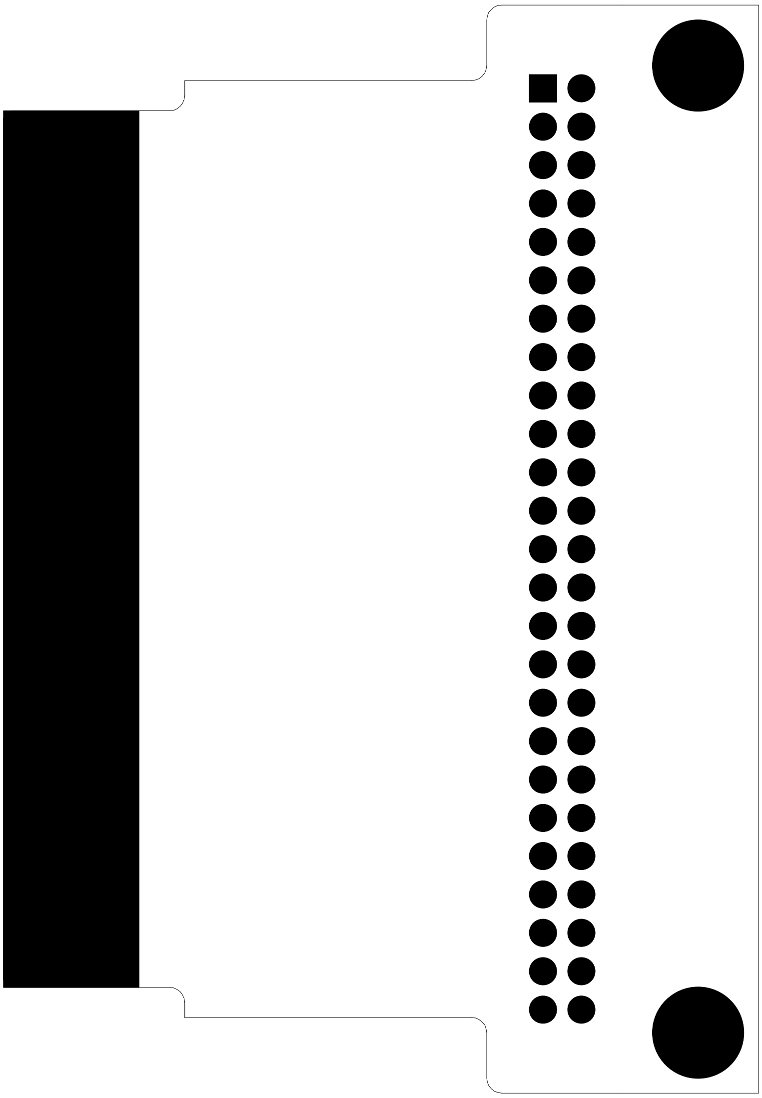
C64_Exp_port		
25.05.2026 17:58		Rev.: 0
stopmask component side		



Sven Petersen 2026	Doc.-No.: 226-2-01-00	
	Cu: 35µm	Cu-Layers: 2
C64_Exp_port		
25.05.2026 17:58		Rev.: 0
stopmask solder side		



Sven Petersen 2026	Doc.-No.: 226-2-01-00	
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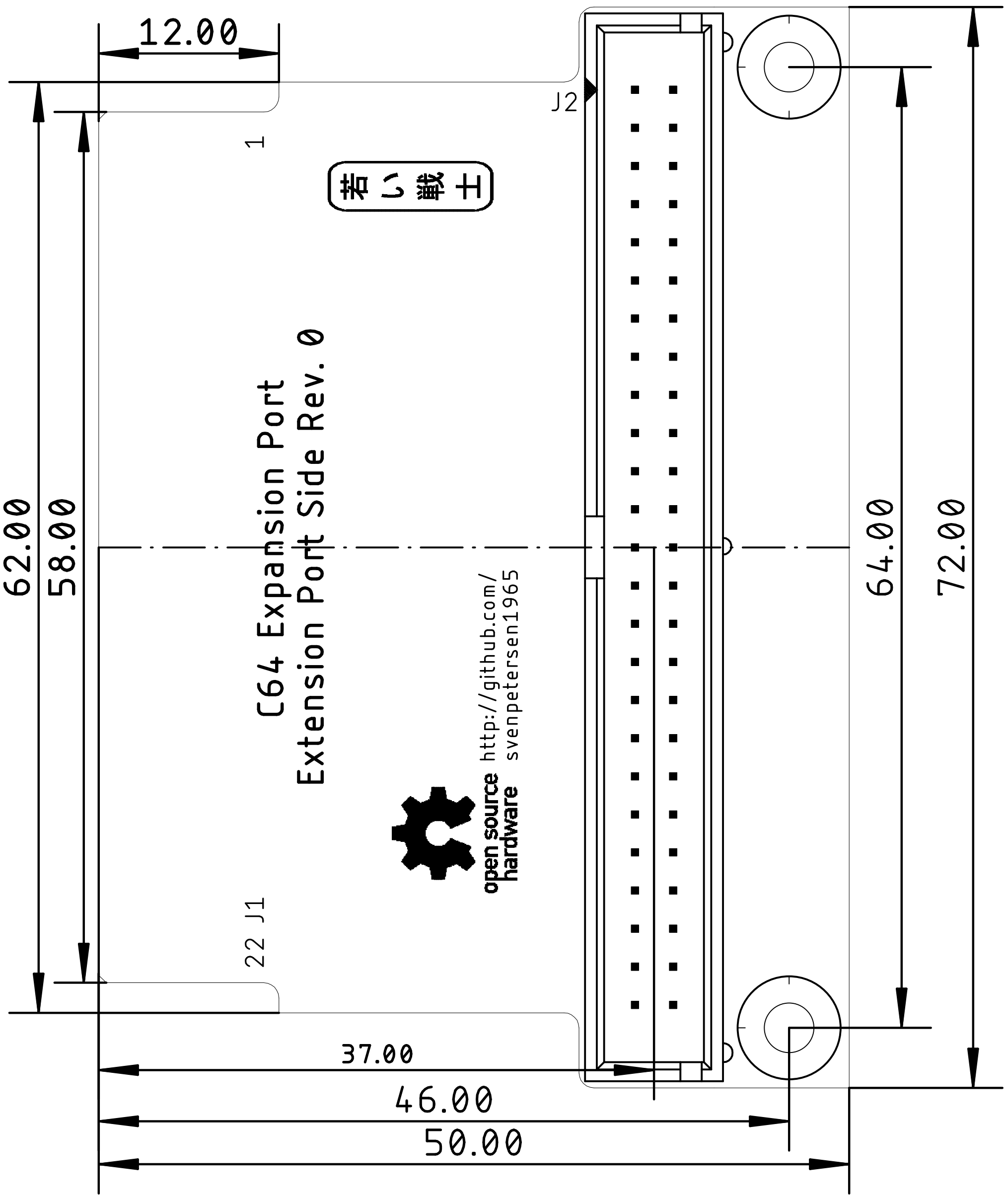
C64_Exp_port		
25.05.2026 17:58		Rev.: 0
placement component side measures		



C64 Expansion Port Extension Port Side Rev. 0
Bill of Material Rev. 0.0

Pos.	Qty	Value	Footprint	Ref.-No.	Comment
1	1	226-2-01-00	2 Layer	PCB Rev. 0	2 layer, Cu 35μ, HASL, 72.0mm x 50.0mm, 1.6mm FR4
	1	2x25 box header, 2.54mm	2X25WV	J2	2x25, boxed pin header, e.g. reichelt.de WSL 50G